Multilinear Discriminant Analysis for Face Recognition

Paper-proximate study · original-resolution ORL faces

COMP3314 · Group 27 · Fall 2026
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Ten captures of one ORL identity

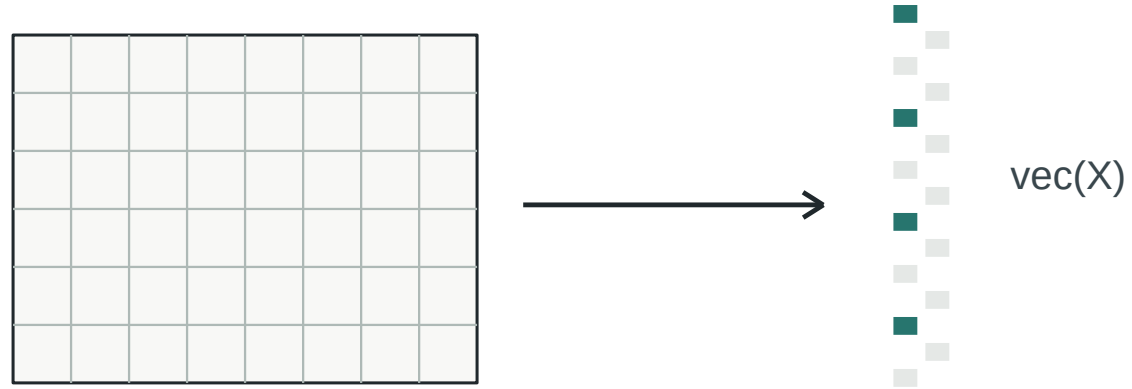
Yan et al. (2007), Fig. 4 · images credited to AT&T Laboratories Cambridge

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Flattening keeps the pixels, but hides the image axes

A 112 × 92 face has 10,304 pixel values.

A vector keeps those values. It no longer presents rows and columns as separate structure to the learner.



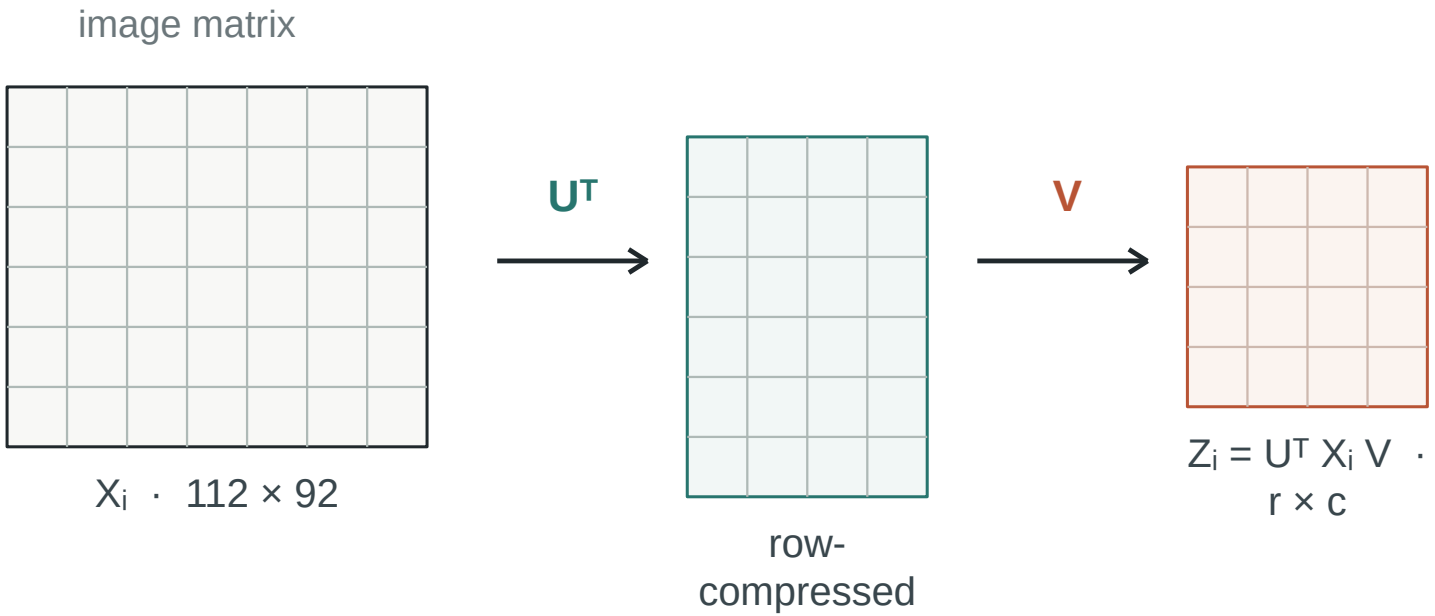
matrix X · rows × columns

PCA keeps directions of high variance. LDA uses identity labels to separate classes.

MDA asks whether those labels work better when row and column modes stay explicit.

Course links: PCA · LDA · high-dimensional feature vectors

MDA projects both axes into a compact face feature



U reduces the row mode
V reduces the column mode.

MDA/2-1 learns one mode.
MDA/2-2 learns both, jointly.

The feature remains a matrix until classification.

Yan et al. call the coupled projections interrelated discriminative subspaces.

The DTC is Fisher's ratio in projected tensor space

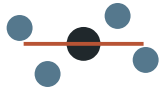
$$J(U,V) = \frac{\sum_c n_c \left\| U^\top (\bar{X}_c - \bar{X}) V \right\|_F^2}{\sum_c \sum_{i:y_i=c} \left\| U^\top (X_i - \bar{X}_c) V \right\|_F^2}$$

between identities



identity means move apart

within one identity



samples stay near their class mean

Like LDA: maximize between-class scatter and minimize within-class scatter.

MDA updates one mode at a time

Each update solves a generalized eigenproblem.

Start with U and V

01 **Update U**

Hold V fixed

$$S_B^{(U)} u = \lambda S_W^{(U)} u$$

02 **Update V**

Hold U fixed

$$S_B^{(V)} v = \lambda S_W^{(V)} v$$

03 **Convergence?**

If no, repeat

k-mode loop · row update, column update, repeat

Gallery images only

The learned projections are then applied to gallery and probe features; 1-nearest neighbour classifies each probe.

Original ORL, matched gallery/probe splits

400	40	10	92 × 112
images	identities	images / identity	pixels (W × H)

Example: three gallery images



gallery remaining images become probes

→ **fit U,V on gallery** → **Euclidean 1-NN**

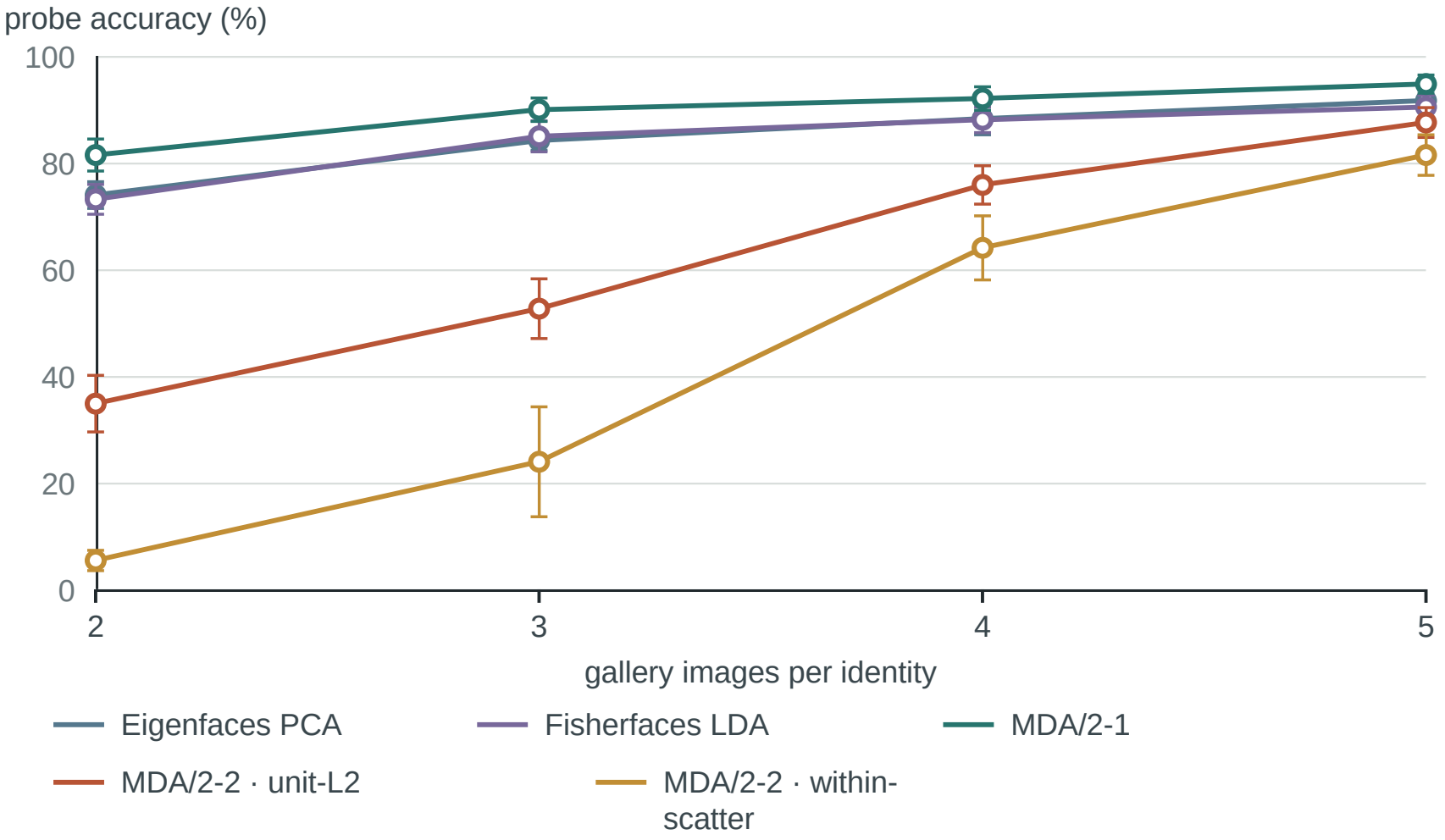
Fixed-setting study: 30 shared splits for each gallery size 2–5. Full dimension grid: five repeats.

Paper says histogram equalization. We use 64 levels; exact equalization defaults and ridge handling are not specified in the paper.

ORL source: AT&T Laboratories Cambridge · image dimensions shown as width × height

MDA/2-1 leads the fixed-setting comparison

MDA/2-2 changes sharply when the projection-axis normalization changes.



6 × 6 MDA/2-2 · same 1-NN classifier

At two gallery images, changing only the axis scaling moves MDA/2-2 from 34.8% to 5.4%.

Means ± sample SD
30 matched splits

Our full-grid peaks are lower than Table I

Protocols differ; our dimension choices use the same probe scores shown here.



10,130

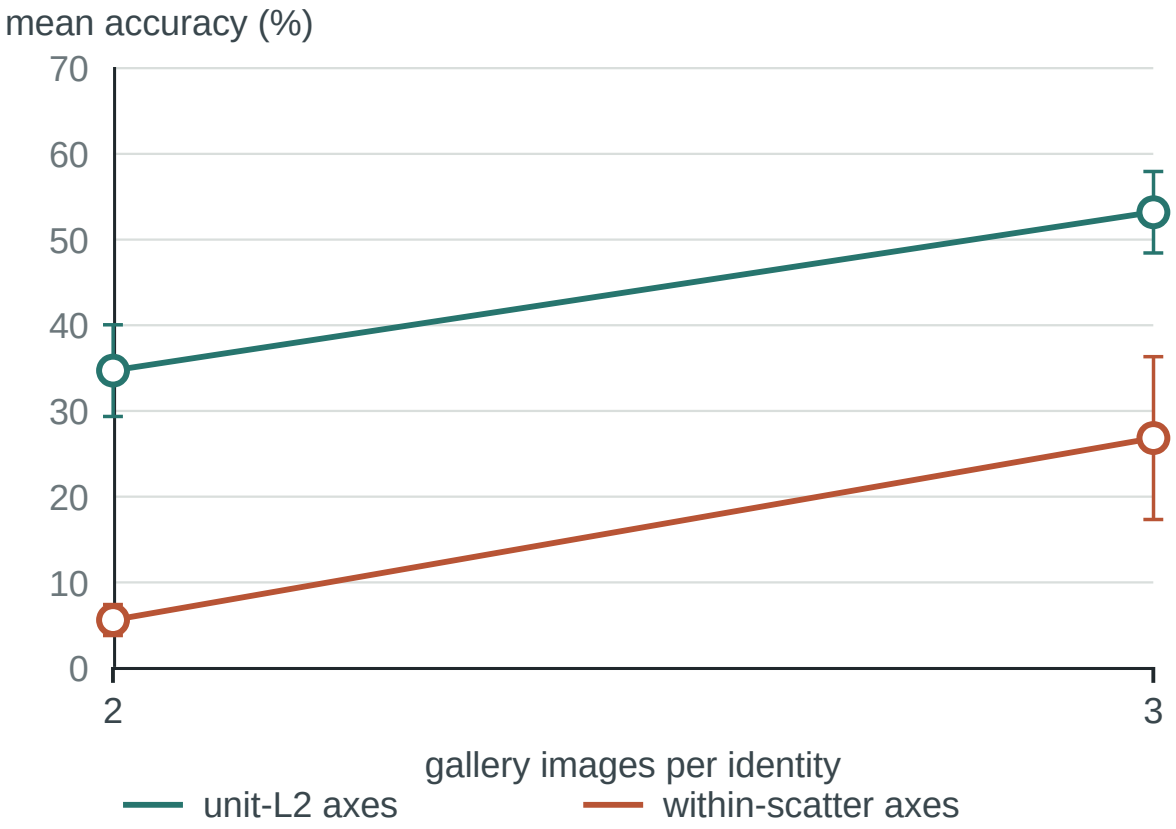
feasible row × column pairs

Five repeats per gallery size and normalization.

The group's best dimensions were selected using the same probe scores shown here. Those maxima are optimistic, not held-out test estimates.

Paper: reported best scores. Group: five-repeat grid maxima. The protocols are not matched.

At fixed 6 × 6 dimensions, a third gallery image helps



✦18.5 pp

unit-L2: 34.57% → 53.05%

✦21.2 pp

within-scatter: 5.47% → 26.70%

30 nested gallery splits · dimensions held at 6 × 6 · error bars show sample SD

Our evidence supports a careful claim

The available paper details do not define an exact numerical reproduction.

What MDA offers

Keeps row and column structure.

Updates smaller mode-specific eigenproblems.

Alternating updates do not guarantee a global optimum.

What our results show

MDA/2-1 leads our fixed-setting runs.

MDA/2-2 is sensitive to axis scaling.

The selected grid peaks are optimistic because probes choose dimensions.

What remains uncertain

ORL is small, grayscale and controlled.

The paper omits split seeds, scaling, singular-scatter handling and full dimension-selection details.

Our choices include 64-level equalization, relative 10^{-6} ridge, $\varepsilon=10^{-4}$ and a 10-iteration cap.

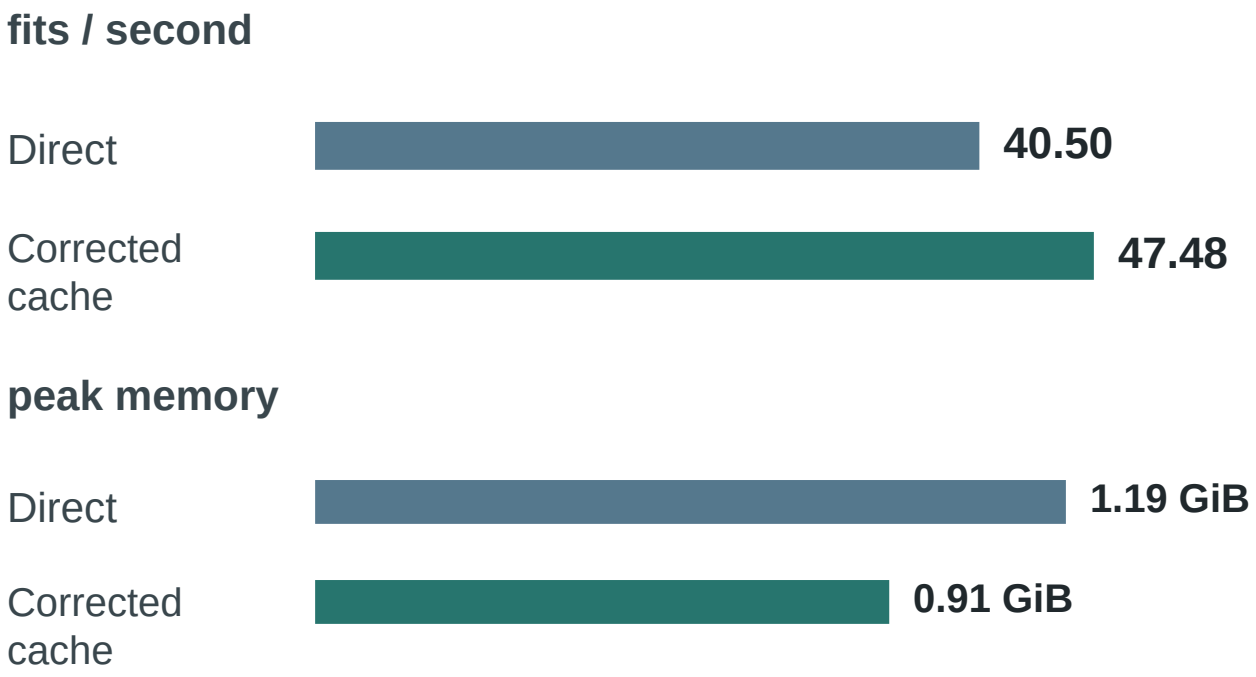
Best next step

Choose dimensions on training-only validation, then report once on an untouched outer test split.

The faster default preserved the completed evidence

A targeted validation avoided repeating the full 202,600-fit grid.

Local M3 Max MacBook Pro · 14 workers · one BLAS thread · 60 fits at G=5



71 min is a batch-based runtime estimate, not a completed full rerun.

Code, protocol and results: Group 27 GitHub repository

17 flagged settings

After the cache correction, direct and optimized fits matched on predictions, accuracy, iterations, projections, distances and objectives under both scalings.

The complete direct-run grids remain canonical. The optimized implementation is now the default for future sweeps.

References and project files

Yan, S. et al. (2007). “Multilinear Discriminant Analysis for Face Recognition.” IEEE Transactions on Image Processing, 16(1), 212–220.

doi.org/10.1109/TIP.2006.884929

AT&T Laboratories Cambridge. The Database of Faces (ORL).

cam-orl.co.uk/facedatabase.html · 40 identities × 10 images · 92 × 112 pixels

COMP3314 lecture links: PCA/LDA and k-nearest neighbours.

Group 27 implementation, run instructions and experimental artifacts

github.com/Lavalx100/comp3314-mda-face-recognition-replication

Questions